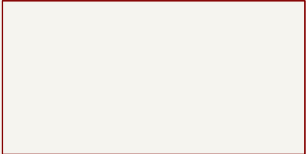


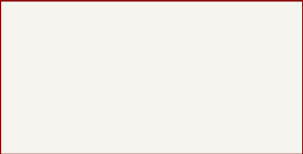
# Data Center RDIMM DDR5 Tester

HyperRAM



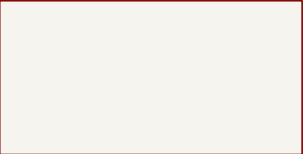
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DDR5



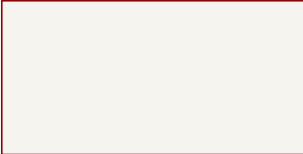
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Interfaces



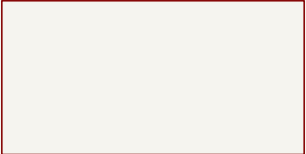
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Ethernet



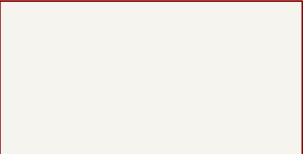
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Supply



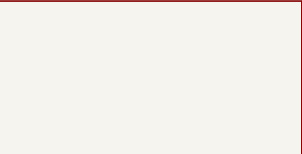
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Config SPI flash



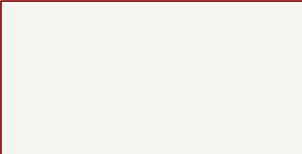
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FPGA power



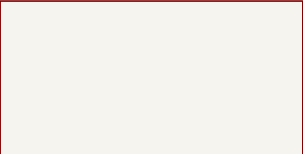
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FPGA banks 12-15



File: fpga-banks-12-15.kicad\_sch

FPGA banks 16-34



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Logo N2 oshw\_logo

Logo N1 antmicro\_logo

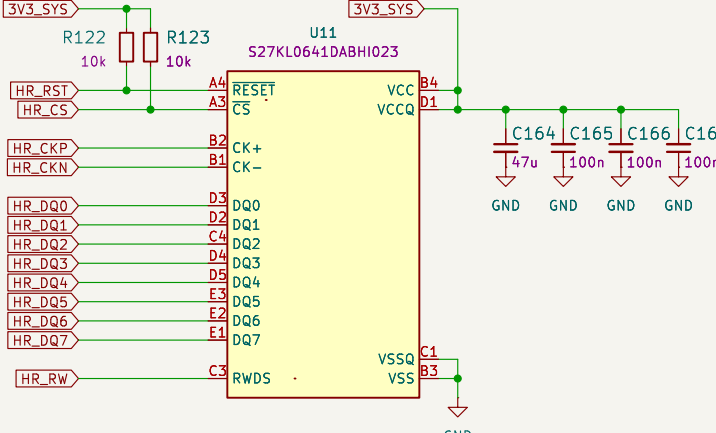
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Sheet: /  
File: data-center-rdim--ddr5-tester.kicad\_sch

**Title: Data Center RDIMM DDR5 Tester**

|                    |                  |                  |
|--------------------|------------------|------------------|
| Size: A3           | Date: 2023-07-27 | Rev: 1.0.1:85273 |
| KiCad E.D.A. 8.0.8 | Id: 1/10         |                  |

# HyperRAM



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Sheet: /HyperRAM/  
File: hyperram.kicad\_sch

# Title: Data Center RDIMM DDR5 Tester

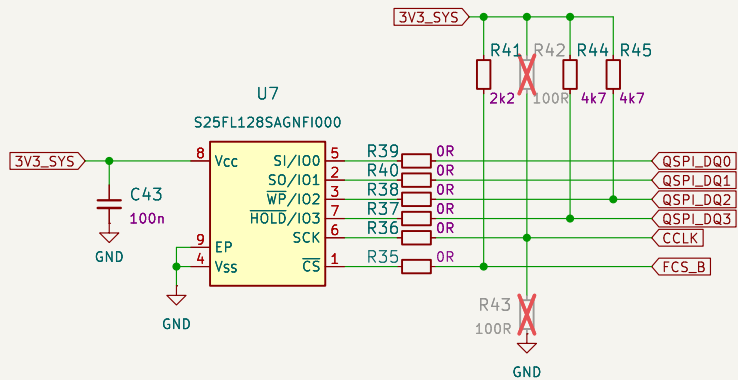
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| Size: A3           | Date: 2023-07-27 |
| KiCad E.D.A. 8.0.8 |                  |

Rev: 1.0.1:85273  
Id: 2/10

Master SPI Quad (x4) configuration scheme

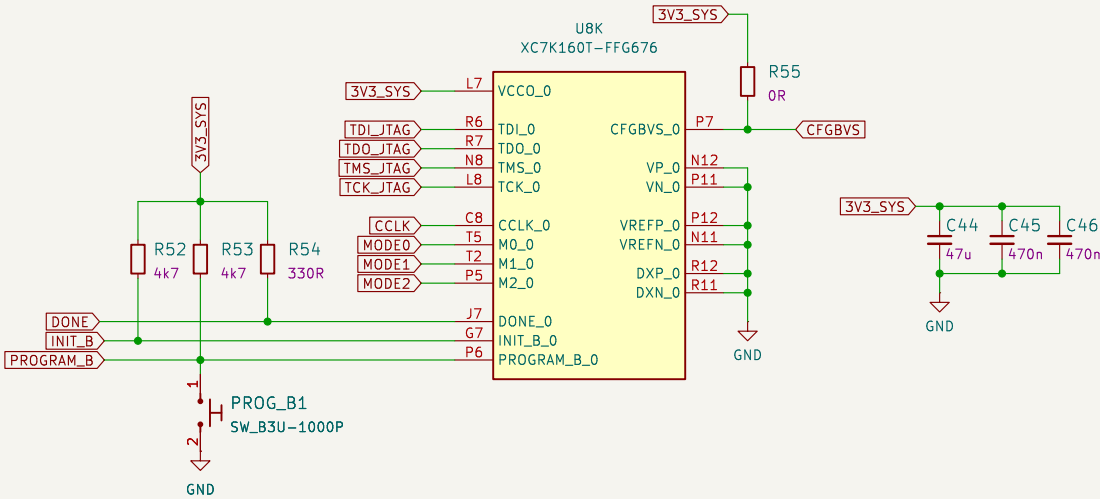
Follows Figure 2-14 7 Series FPGAs Configuration User Guide  
UG470 (v1.13.1)

(Q)SPI flash



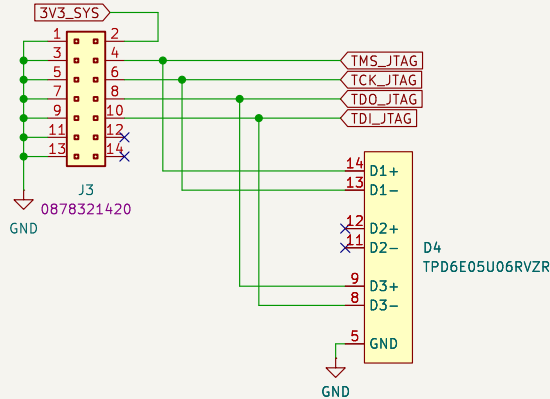
FPGA BANK 0

TODD: verify after FPGA swap

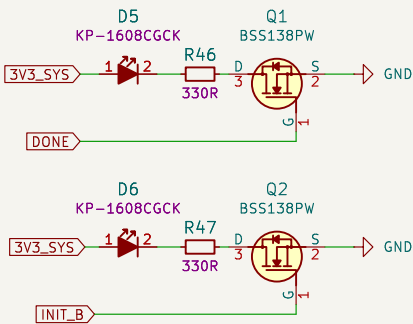


JTAG Connector

Compatible with Xilinx Platform Cable

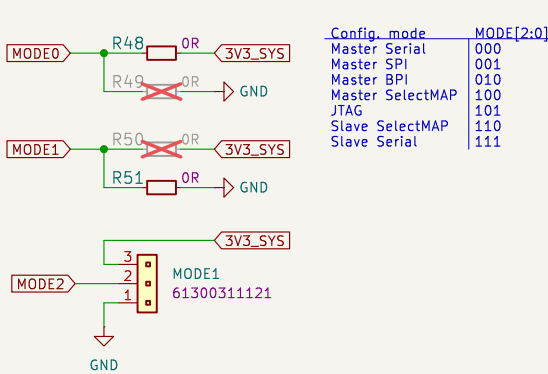


STATUS LEDs

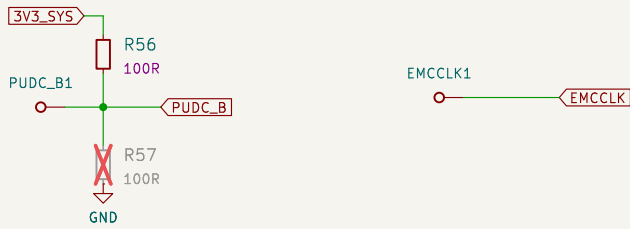


Configuration Modes

For details, see UG470 p. 21



Probes



The diagram illustrates the internal architecture and pin connections of the DDR504111002KQ memory module. It is divided into two main sections: U12A (left) and U12B (right).

**U12A (Left Section):**

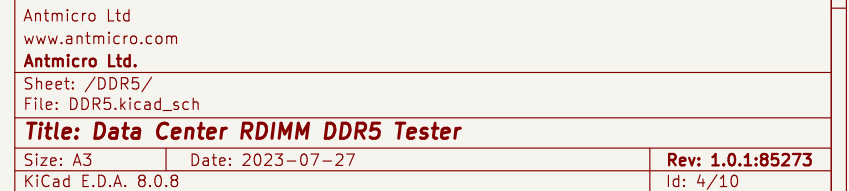
- Inputs:** DQ0\_A, DQ1\_A, DQ2\_A, DQ3\_A, DQ50\_A\_P, DQ50\_A\_N, DQ4\_A, DQ5\_A, DQ6\_A, DQ7\_A, DQ55\_A\_P, DQ55\_A\_N, DQ8\_A, DQ9\_A, DQ10\_A, DQ11\_A, DQ51\_A\_P, DQ51\_A\_N, DQ12\_A, DQ13\_A, DQ14\_A, DQ15\_A, DQ56\_A\_P, DQ56\_A\_N, DQ16\_A, DQ17\_A, DQ18\_A, DQ19\_A, DQ52\_A\_P, DQ52\_A\_N, DQ20\_A, DQ21\_A, DQ22\_A, DQ23\_A, DQ57\_A\_P, DQ57\_A\_N, DQ24\_A, DQ25\_A, DQ26\_A, DQ27\_A, DQ53\_A\_P, DQ53\_A\_N, DQ28\_A, DQ29\_A, DQ30\_A, DQ31\_A, DQ58\_A\_P, DQ58\_A\_N, CB0\_A, CB1\_A, CB2\_A, CB3\_A, DQ54\_A\_P, DQ54\_A\_N, CB4\_A, CB5\_A, CB6\_A, CB7\_A, DQ59\_A\_P, DQ59\_A\_N.
- Outputs:** DQ0\_B, DQ1\_B, DQ2\_B, DQ3\_B, DQ50\_B\_P, DQ50\_B\_N, DQ4\_B, DQ5\_B, DQ6\_B, DQ7\_B, DQ55\_B\_P, DQ55\_B\_N, DQ8\_B, DQ9\_B, DQ10\_B, DQ11\_B, DQ51\_B\_P, DQ51\_B\_N, DQ12\_B, DQ13\_B, DQ14\_B, DQ15\_B, DQ56\_B\_P, DQ56\_B\_N, DQ16\_B, DQ17\_B, DQ18\_B, DQ19\_B, DQ52\_B\_P, DQ52\_B\_N, DQ20\_B, DQ21\_B, DQ22\_B, DQ23\_B, DQ57\_B\_P, DQ57\_B\_N, DQ24\_B, DQ25\_B, DQ26\_B, DQ27\_B, DQ53\_B\_P, DQ53\_B\_N, DQ28\_B, DQ29\_B, DQ30\_B, DQ31\_B, DQ58\_B\_P, DQ58\_B\_N, CB0\_B, CB1\_B, CB2\_B, CB3\_B, DQ54\_B\_P, DQ54\_B\_N, CB4\_B, CB5\_B, CB6\_B, CB7\_B, DQ59\_B\_P, DQ59\_B\_N.
- Internal Connections:** DQ0\_A to DQ0\_B, DQ1\_A to DQ1\_B, DQ2\_A to DQ2\_B, DQ3\_A to DQ3\_B, DQ50\_A\_P to DQ50\_B\_P, DQ50\_A\_N to DQ50\_B\_N, DQ4\_A to DQ4\_B, DQ5\_A to DQ5\_B, DQ6\_A to DQ6\_B, DQ7\_A to DQ7\_B, DQ55\_A\_P to DQ55\_B\_P, DQ55\_A\_N to DQ55\_B\_N, DQ8\_A to DQ8\_B, DQ9\_A to DQ9\_B, DQ10\_A to DQ10\_B, DQ11\_A to DQ11\_B, DQ51\_A\_P to DQ51\_B\_P, DQ51\_A\_N to DQ51\_B\_N, DQ12\_A to DQ12\_B, DQ13\_A to DQ13\_B, DQ14\_A to DQ14\_B, DQ15\_A to DQ15\_B, DQ56\_A\_P to DQ56\_B\_P, DQ56\_A\_N to DQ56\_B\_N, DQ16\_A to DQ16\_B, DQ17\_A to DQ17\_B, DQ18\_A to DQ18\_B, DQ19\_A to DQ19\_B, DQ52\_A\_P to DQ52\_B\_P, DQ52\_A\_N to DQ52\_B\_N, DQ20\_A to DQ20\_B, DQ21\_A to DQ21\_B, DQ22\_A to DQ22\_B, DQ23\_A to DQ23\_B, DQ57\_A\_P to DQ57\_B\_P, DQ57\_A\_N to DQ57\_B\_N, DQ24\_A to DQ24\_B, DQ25\_A to DQ25\_B, DQ26\_A to DQ26\_B, DQ27\_A to DQ27\_B, DQ53\_A\_P to DQ53\_B\_P, DQ53\_A\_N to DQ53\_B\_N, DQ28\_A to DQ28\_B, DQ29\_A to DQ29\_B, DQ30\_A to DQ30\_B, DQ31\_A to DQ31\_B, DQ58\_A\_P to DQ58\_B\_P, DQ58\_A\_N to DQ58\_B\_N, CB0\_A to CB0\_B, CB1\_A to CB1\_B, CB2\_A to CB2\_B, CB3\_A to CB3\_B, DQ54\_A\_P to DQ54\_B\_P, DQ54\_A\_N to DQ54\_B\_N, CB4\_A to CB4\_B, CB5\_A to CB5\_B, CB6\_A to CB6\_B, CB7\_A to CB7\_B, DQ59\_A\_P to DQ59\_B\_P, DQ59\_A\_N to DQ59\_B\_N.

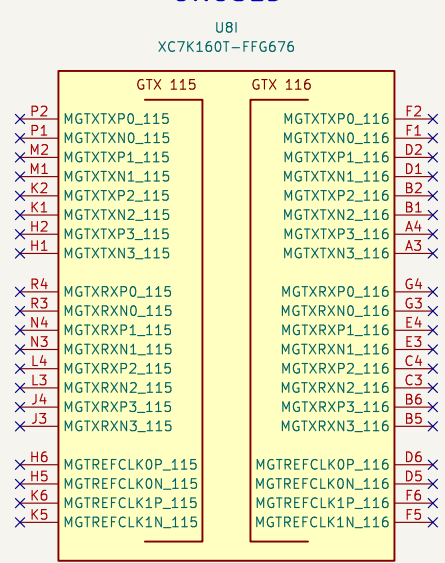
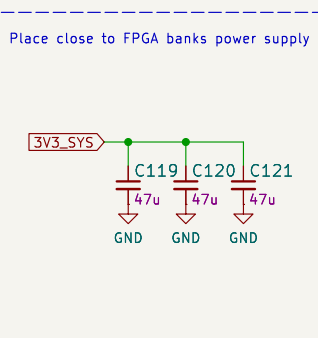
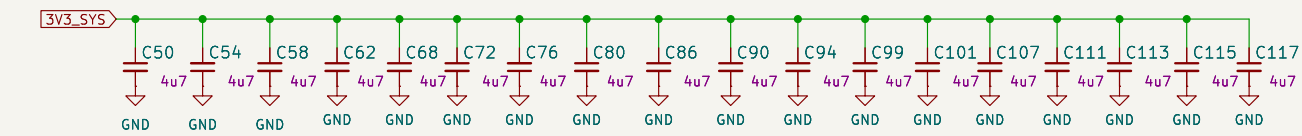
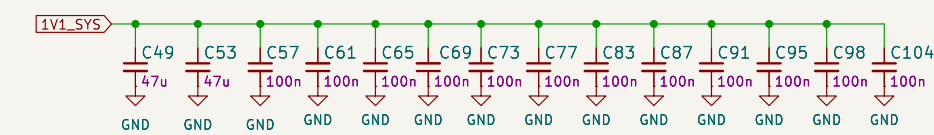
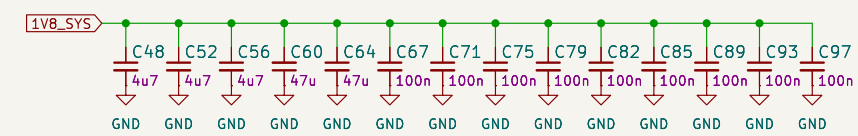
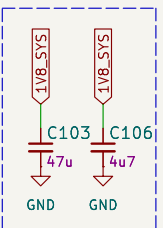
**U12B (Right Section):**

- Inputs:** VIN\_BULK, VIN\_MGMT, PCAMP, DLBDQ, DLBDQS, RESET\_N, CA0\_A, CA1\_A, CA2\_A, CA3\_A, CA4\_A, CA5\_A, CA6\_A, PAR\_A, CS0\_A\_N, CS1\_A\_N.
- Outputs:** HSA, HSDA, HSCL, ALERT\_N, CK\_T, CK\_C, CA0\_B, CA1\_B, CA2\_B, CA3\_B, CA4\_B, CA5\_B, CA6\_B, PAR\_B, CS0\_B\_N, CS1\_B\_N.
- Internal Connections:** VIN\_BULK to HSA, VIN\_MGMT to HSDA, PCAMP to HSCL, DLBDQ to ALERT\_N, DLBDQS to CK\_T, RESET\_N to CK\_C, CA0\_A to CA0\_B, CA1\_A to CA1\_B, CA2\_A to CA2\_B, CA3\_A to CA3\_B, CA4\_A to CA4\_B, CA5\_A to CA5\_B, CA6\_A to CA6\_B, PAR\_A to PAR\_B, CS0\_A\_N to CS0\_B\_N, CS1\_A\_N to CS1\_B\_N.

**Pin Connections:**

- U12A:** DQ0\_A, DQ1\_A, DQ2\_A, DQ3\_A, DQ50\_A\_P, DQ50\_A\_N, DQ4\_A, DQ5\_A, DQ6\_A, DQ7\_A, DQ55\_A\_P, DQ55\_A\_N, DQ8\_A, DQ9\_A, DQ10\_A, DQ11\_A, DQ51\_A\_P, DQ51\_A\_N, DQ12\_A, DQ13\_A, DQ14\_A, DQ15\_A, DQ56\_A\_P, DQ56\_A\_N, DQ16\_A, DQ17\_A, DQ18\_A, DQ19\_A, DQ52\_A\_P, DQ52\_A\_N, DQ20\_A, DQ21\_A, DQ22\_A, DQ23\_A, DQ57\_A\_P, DQ57\_A\_N, DQ24\_A, DQ25\_A, DQ26\_A, DQ27\_A, DQ53\_A\_P, DQ53\_A\_N, DQ28\_A, DQ29\_A, DQ30\_A, DQ31\_A, DQ58\_A\_P, DQ58\_A\_N, CB0\_A, CB1\_A, CB2\_A, CB3\_A, DQ54\_A\_P, DQ54\_A\_N, CB4\_A, CB5\_A, CB6\_A, CB7\_A, DQ59\_A\_P, DQ59\_A\_N.
- U12B:** VIN\_BULK, VIN\_MGMT, PCAMP, DLBDQ, DLBDQS, RESET\_N, CA0\_A, CA1\_A, CA2\_A, CA3\_A, CA4\_A, CA5\_A, CA6\_A, PAR\_A, CS0\_A\_N, CS1\_A\_N.

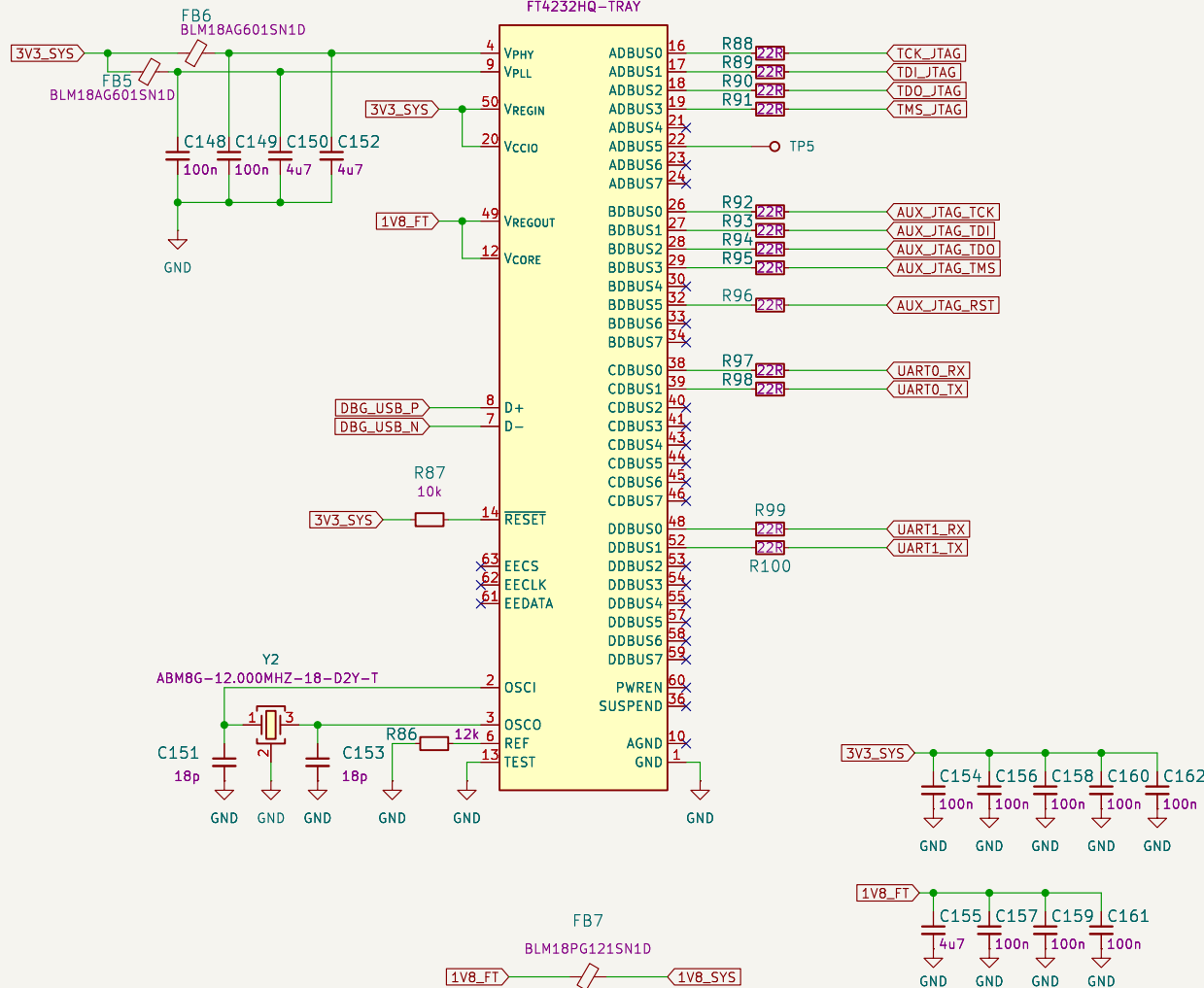




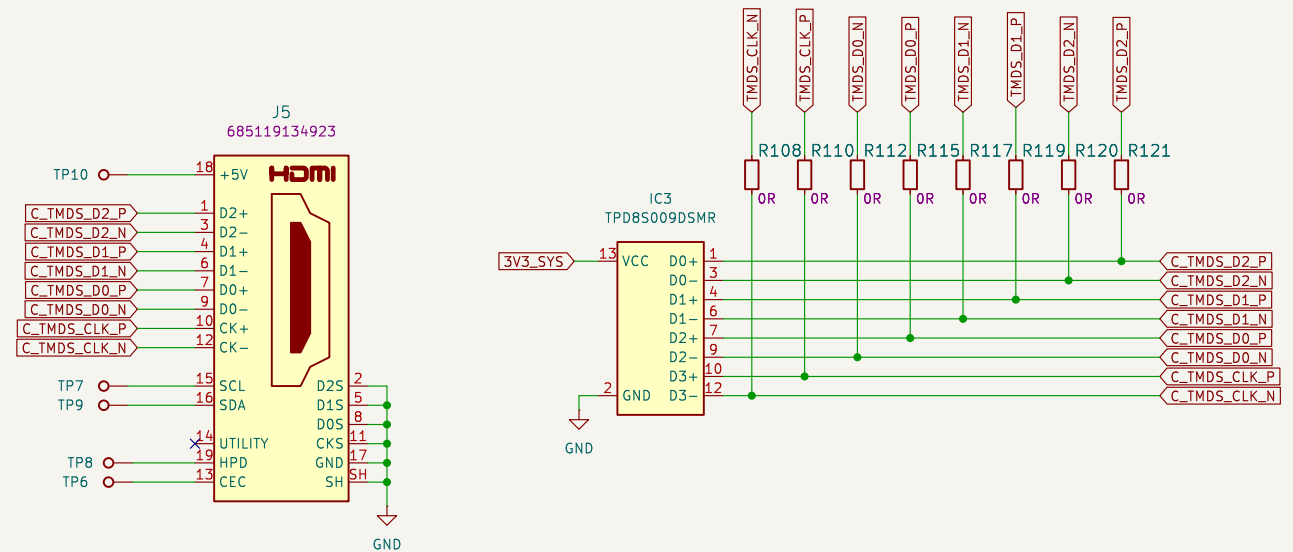
## Debug UART

U10

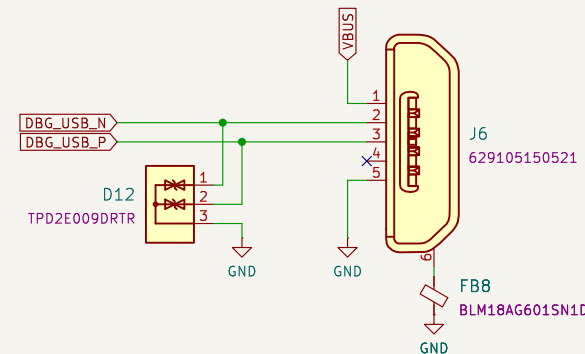
FT4232HQ-TRAY



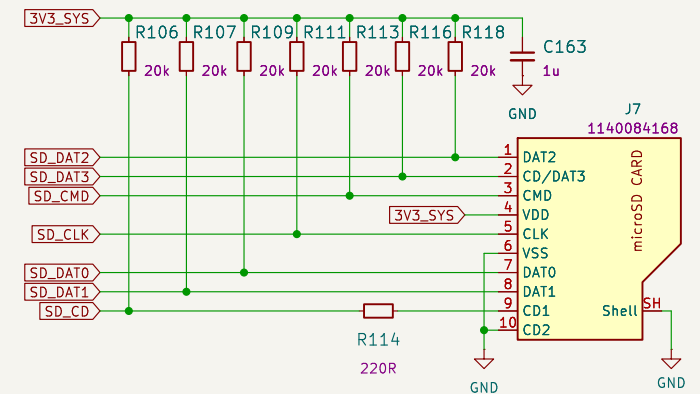
## HDMI



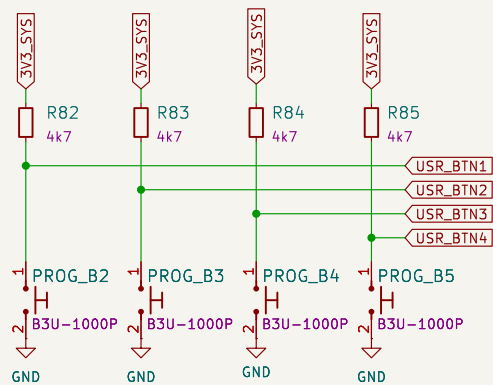
## Debug USB



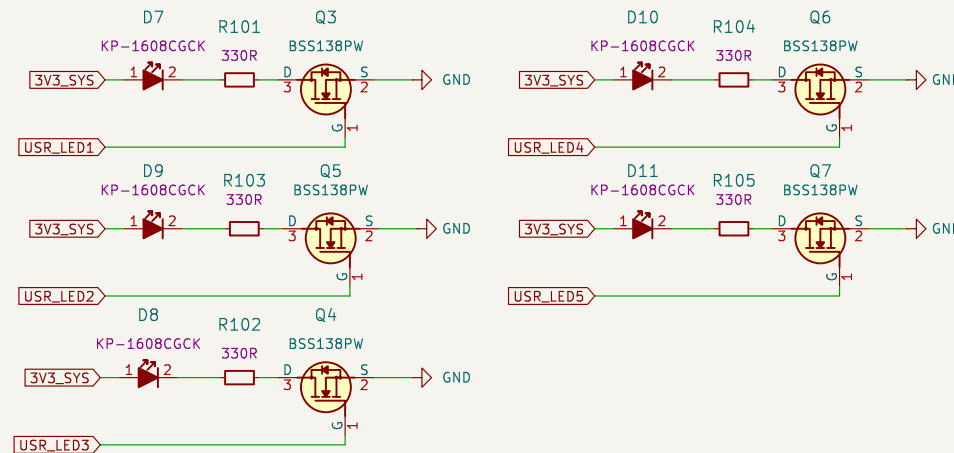
## SD card



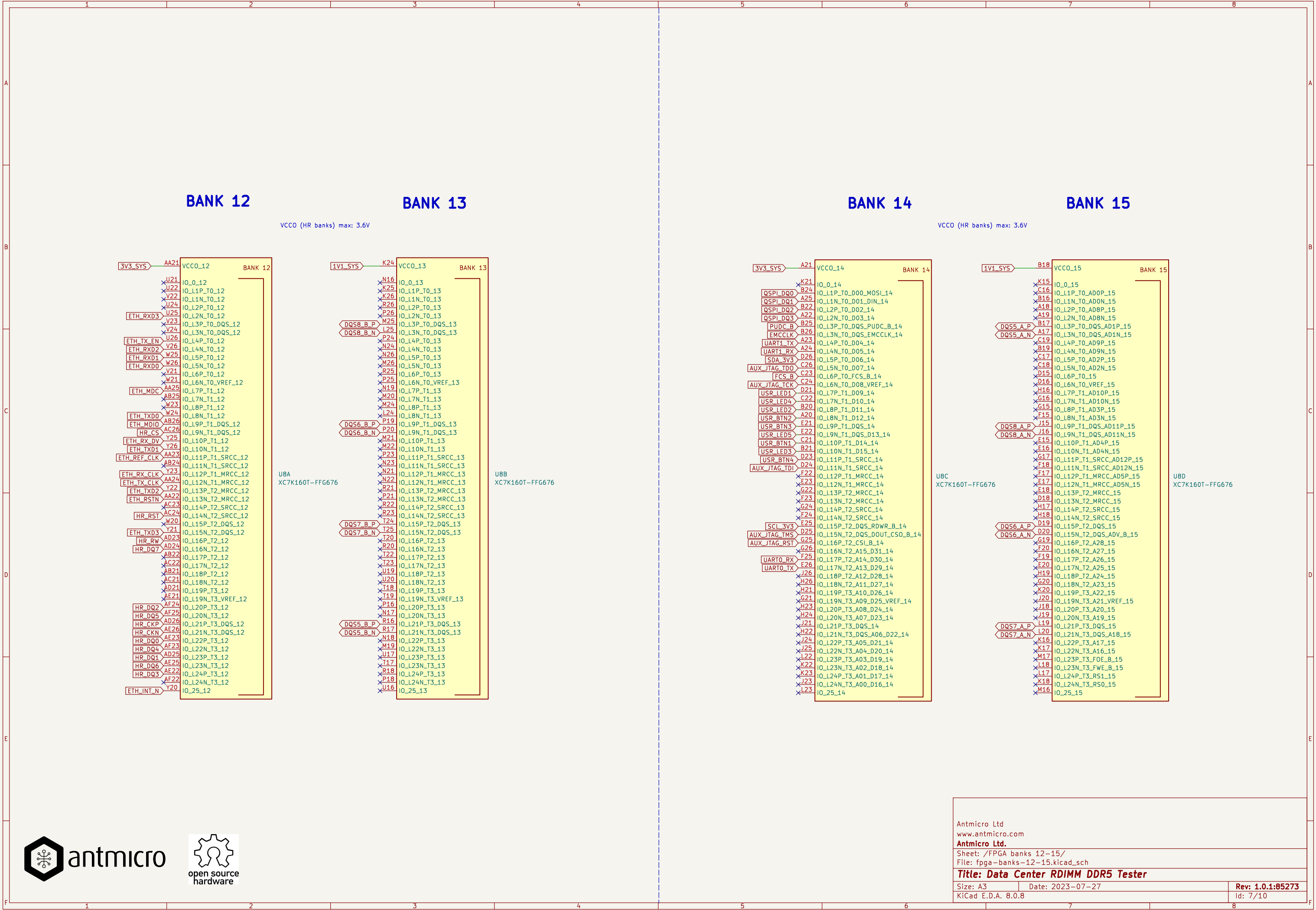
## User buttons



## User LEDs







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Title: Data Center RDIMM DDR5 Tester

Size: A3

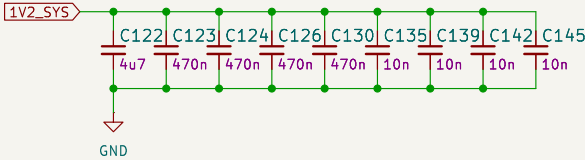
Date: 2023-07-27

Rev: 1.0.1:85273

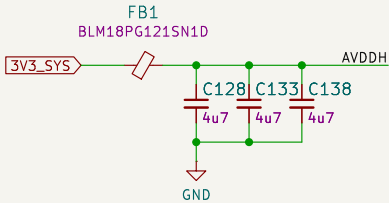
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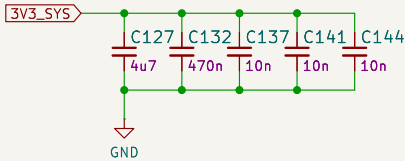
DVDDL decoupling



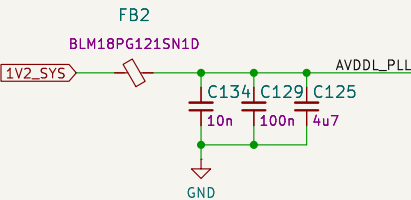
AVDDH decoupling



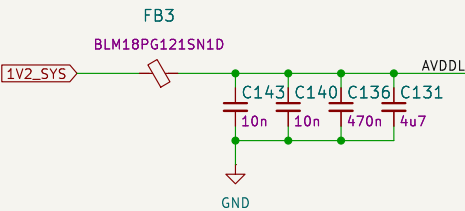
DVDDH decoupling



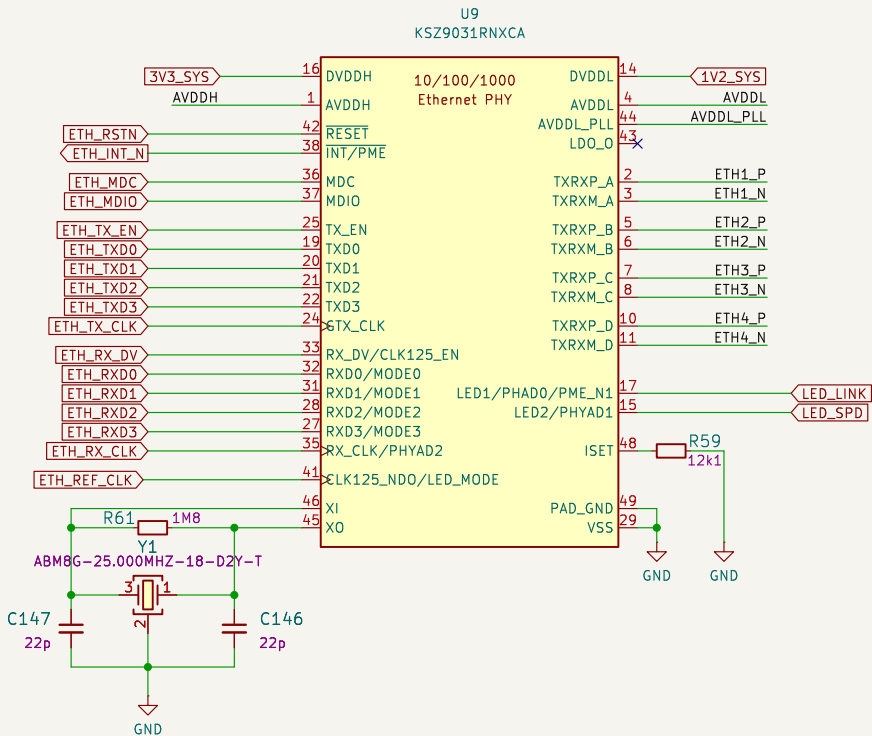
AVDDL\_PLL decoupling



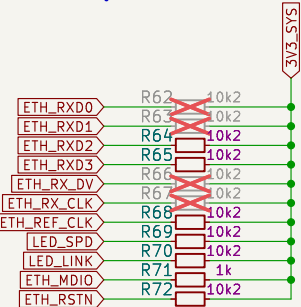
AVDDL decoupling



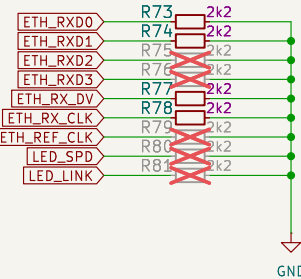
PHY



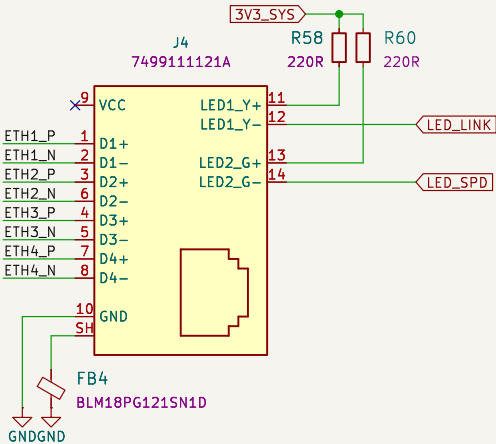
Pull up resistors



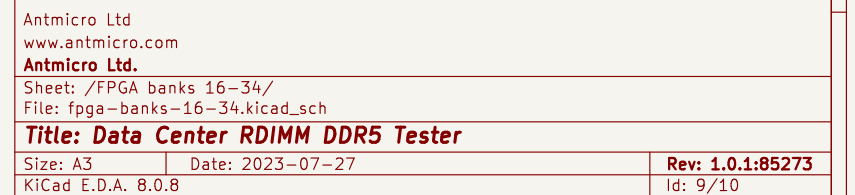
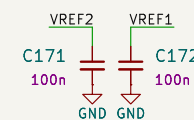
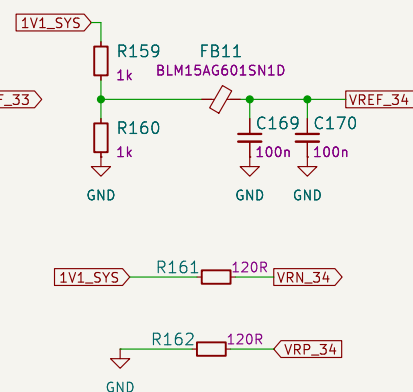
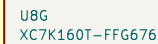
Pull down resistors



RJ45 Connector



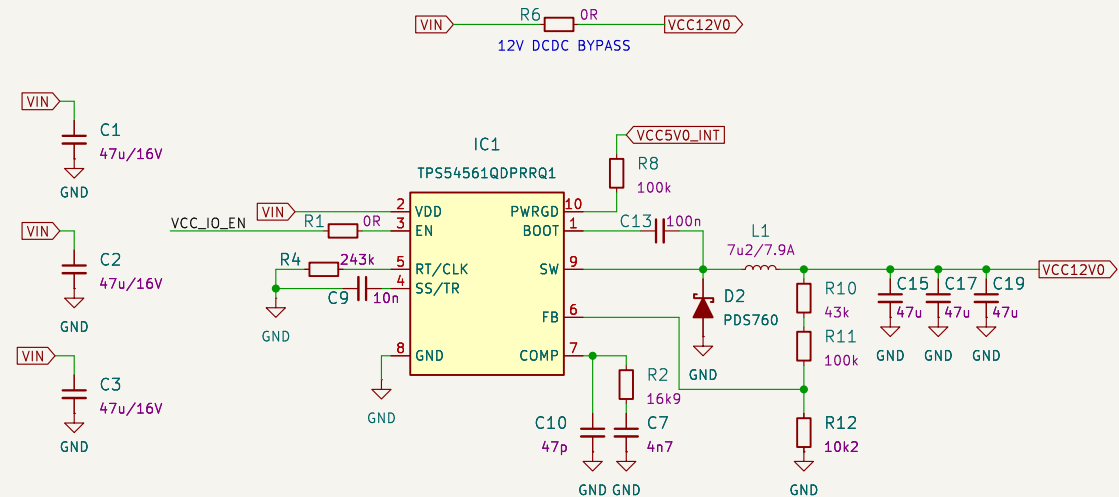




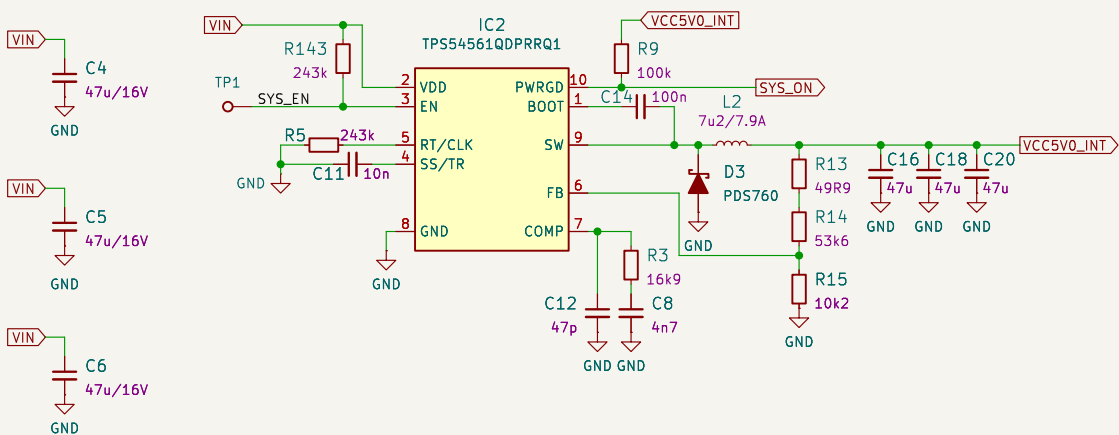
Input power connector



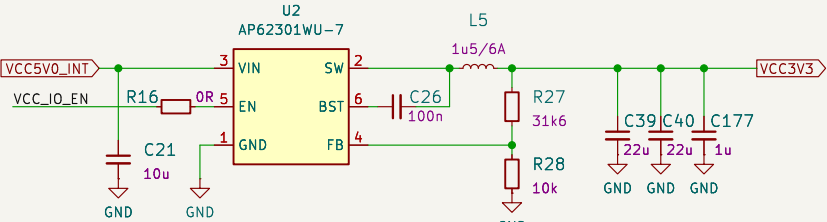
Main supply (12V 5A)



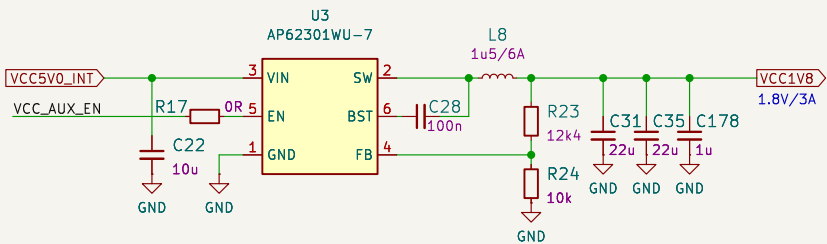
5V0 supply



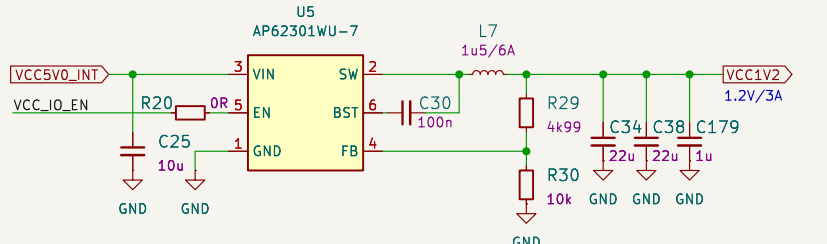
3V3 supply (3A)



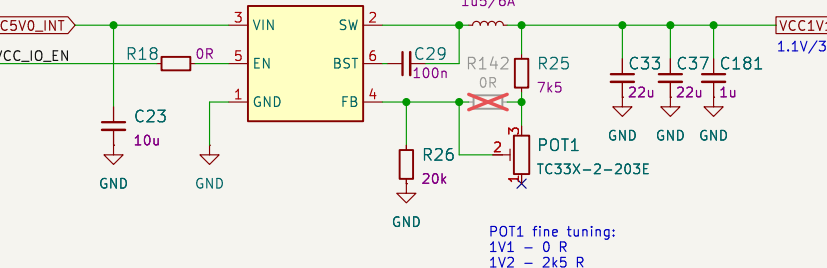
1V8 supply



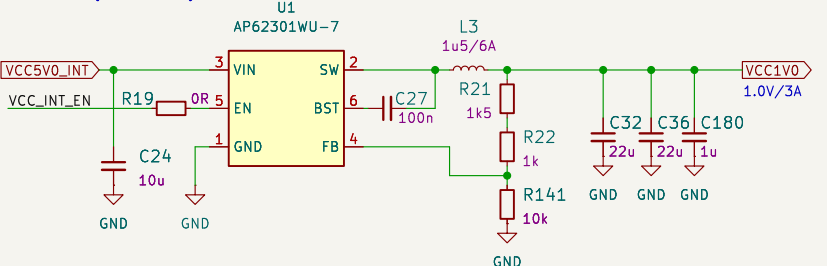
1V2 supply



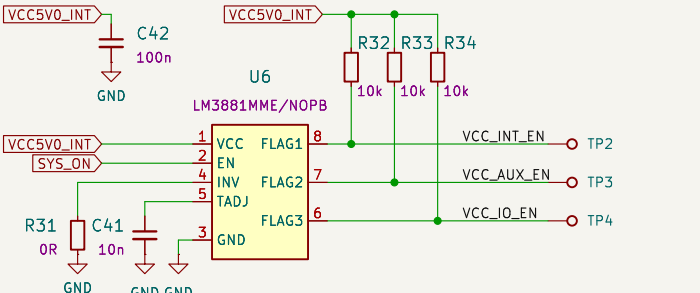
1V1 supply



VCCINT (1.0V 3A)

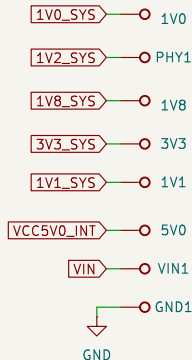


Power sequencer

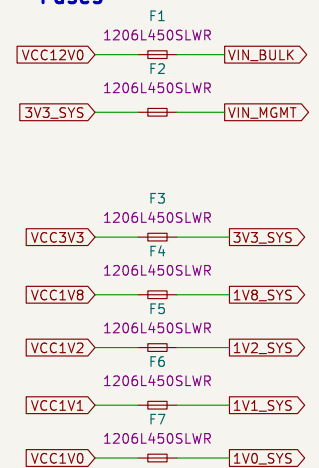


STEP1 - VCCINT (1.0V) for FPGA  
STEP2 - VCCAUX (1.8V, 2.5V, 1.2V) for FPGA and DDR  
STEP3 - VCCIO (3.3V, 1.2V, 0.6V) for FPGA, PHY and DDR

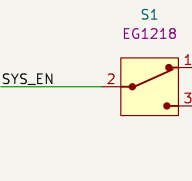
Probes



Fuses



Power switch



Optional FAN connector

